

## THE DISTRIBUTIONS OF SOME TEST CRITERIA DEPENDING ON MULTIVARIATE DIRICHLET DISTRIBUTIONS\*

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### 1. Introduction and Summary

Let  $A_j$  ( $j=1, \dots, k$ ) and  $B$  be independent random matrices having Wishart distributions with the same covariance matrix  $\Sigma$  and  $m_j$  ( $j=1, \dots, k$ ) and  $n$  degrees of freedom, respectively.

Let  $L_j$  be defined by

$$1.1 \quad L_j = T^{-1/2} A_j T^{-1/2}, j=1, \dots, k$$

where  $\sum_1^k A_j + B = T$ , and  $T^{1/2}$  is the square root of  $T$ . We also adopt the convention that in an expression like  $T^{1/2} A T^{1/2}$  the post-multiplier is  $(T^{1/2})'$ .

The joint distribution of  $(L_1, \dots, L_k)$  is the multivariate Dirichlet Type 1 distribution  $MD_1(\frac{1}{2}m_1, \frac{1}{2}m_2, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$  with density function (Olkin and Rubin (1964), Troskie (1966, 1967))

$$1.2 \quad D(m_1, \dots, m_k; n) \prod_1^k |L_j|^{1/2(m_j - p - 1)} |I - \sum_1^k L_j|^{1/2(n - p - 1)}, L_j > 0, I - \sum_1^k L_j > 0$$

where

$$1.3 \quad D(m_1, \dots, m_k; n) = \Gamma_p(\frac{1}{2}(m+n)) / \Gamma_p(\frac{1}{2}n) \prod_1^k \Gamma_p(\frac{1}{2}m_j) \text{ and } m = \sum_1^k m_j.$$

$A > 0$  means that  $A$  is positive definite and  $\Gamma_p(\delta) = \pi^{p/2} p(p-1) \Gamma(\delta) \cdot \Gamma(\delta - \frac{1}{2}) \dots \Gamma(\delta - \frac{1}{2}(p-1))$  is the multivariate Gamma function.

Similarly the joint distribution of  $(V_1, \dots, V_k)$  where

$$1.4 \quad V_j = B^{-1/2} A_j B^{-1/2}, j=1, \dots, k$$

is the multivariate Dirichlet Type 2 distribution  $MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n)$  with density function

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$$1.5 \quad D(m_1, \dots, m_k; n) \prod_1^k |V_j|^{1/2(m_j - p - 1)} |I + \sum_1^k V_j|^{-1/2(m+n)}, V_j > 0$$

There is the following direct relationship between (1.2) and (1.5). If  $(L_1, \dots, L_k) \sim MD_1(1/2 m_1, \dots, 1/2 m_k; 1/2 n)$  and if

$$1.6 \quad V_j = (I - \sum_1^k L_j)^{-1/2} L_j (I - \sum_1^k L_j)^{-1/2}, j=1, \dots, k$$

then  $(V_1, \dots, V_k) \sim MD_2(1/2 m_1, \dots, 1/2 m_k; 1/2 n)$  (Tan (1969)). Also if  $(V_1, \dots, V_k) \sim MD_2(1/2 m_1, \dots, 1/2 m_k; 1/2 n)$  and

$$1.7 \quad L_j = (I + \sum_1^k V_j)^{-1/2} V_j (I + \sum_1^k V_j)^{-1/2}, j=1, \dots, k$$

then  $(L_1, \dots, L_k) \sim MD_1(1/2 m_1, \dots, 1/2 m_k; 1/2 n)$  (Tan (1969)).

If  $B$  has a noncentral Wishart distribution  $W(\Sigma, \Omega, n)$  then we have noncentral multivariate Dirichlet distributions. The density function of the noncentral multivariate Dirichlet Type 1 distribution  $MD_1(1/2 m_1, \dots, 1/2 m_k; 1/2 n; \Omega)$  is (de Waal (1968))

$$1.8 \quad (1/\Gamma_p(1/2 n)) \prod_1^k \Gamma_p(1/2 m_j) |2\Sigma|^{-1/2(m+n)} \text{etr}(-1/2 \Omega) \prod_1^k |L_j|^{1/2(m_j - p - 1)} \\ |I - \sum_1^k L_j|^{1/2(n - p - 1)} \int_{T > 0} \text{etr}(-1/2 \Sigma^{-1} T) |T|^{1/2(m+n-p-1)} \\ {}_0F_1(1/2 n; \Omega \Sigma^{-1} T^{1/2} (I - \sum_1^k L_j) T^{1/2}) dT$$

Using (1.6) the density of the noncentral multivariate Dirichlet Type 2 distribution  $MD_2(1/2 m_1, \dots, 1/2 m_k; 1/2 n; \Omega)$  is

$$(J(L_1, \dots, L_k \rightarrow V_1, \dots, V_k) = |I + \sum_1^k V_j|^{-1/2(k+1)(p+1)}) \\ 1.9 \quad (1/\Gamma_p(1/2 n)) \prod_1^k \Gamma_p(1/2 m_j) |2\Sigma|^{-1/2(m+n)} \text{etr}(-1/2 \Omega) \prod_1^k |V_j|^{1/2(m_j - p - 1)} \\ |I + \sum_1^k V_j|^{-1/2(m+n)} \int_{T > 0} \text{etr}(-1/2 \Sigma^{-1} T) |T|^{1/2(m+n-p-1)} \\ {}_0F_1(1/2 n; \Omega \Sigma^{-1} T^{1/2} (I + \sum_1^k V_j) T^{1/2}) dT$$

If in (1.4)  $\Sigma = I$ , then (1.9) can be simplified as (de Waal (1968))

$$1.10 \quad D(m_1, \dots, m_k; 1/2 n) \text{etr}(-1/2 \Omega) \prod_1^k |V_j|^{1/2(m_j - p - 1)} |I + \sum_1^k V_j|^{-1/2(m+n)} \\ {}_1F_1(1/2(m+n); 1/2 n; \Omega (I + \sum_1^k V_j)^{-1}), V_j > 0$$

The multivariate Dirichlet distributions and the multivariate Beta distributions arise in various problems in multivariate analysis. Various test criteria proposed in the literature for testing hypotheses in multivariate analysis depend on these distributions. If  $k=1$ , i.e.  $L=T^{-1/2}AT^{-1/2}$  has a multivariate Beta distribution then these test criteria are (Pillai (1955), Troskie (1971a, 1971c))  $|L|, |I-L|, \text{tr}L, \text{tr}L(I-L)^{-1}, \text{tr}L^{-1}(I-L)$  and the largest root of  $L$ . The likelihood ratio test for testing the hypothesis that  $k$  multivariate normal distributions are identical reduces to (Anderson (1958), Troskie (1966))

$$1.11 \quad \Lambda = \prod_1^k |L_j|^{1/2 m_j}$$

As in the case where  $k=1$  other criteria might also be considered for testing the above hypothesis. These are  $|I - \sum_1^k L_j|, \text{tr} \sum_1^k L_j, \text{tr} \sum_1^k L_j (I - \sum_1^k L_j)^{-1}$  and  $\text{tr}(\sum_1^k L_j (I - \sum_1^k L_j))^{-1}$ .

The purpose of this paper is to derive the distributions of the above criteria. The moments and some further extensions of the above criteria have been considered in Troskie (1971b). The asymptotic properties and some power comparisons of these statistics will be considered in a later paper.

## 2. The distributions of the criteria

The following results are of importance.

### *Lemma 2.1*

*If  $R$  and  $S$  are positive definite symmetric matrices of order  $p$  then (Constantine (1963))*

$$2.1 \quad \int_{T>0} \text{etr}(-\frac{1}{2}RT) |T|^{a-\frac{1}{2}(p+1)} C_k(ST) dT$$

$$= \Gamma_p(a) |R|^{-a} {}_K C_K(R^{-1}S),$$

and

$$2.2 \quad \int_{0 < z < I} |Z|^{a-\frac{1}{2}(p+1)} |I-Z|^{b-\frac{1}{2}(p+1)} C_K(RZ) dZ \\ = (\Gamma_p(a) \Gamma_p(b) / \Gamma_p(a+b)) ({}_K C_K(a) / {}_K C_K(a+b)) C_K(R)$$

### *Lemma 2.2*

*If  $\phi(L_1, \dots, L_k) = \prod_1^k |L_j|^{a_j-\frac{1}{2}(p+1)} |I - \sum_1^k L_j|^{b-\frac{1}{2}(p+1)}$  then (Tan (1969))*

$$2.3 \quad \int_{Z=\Sigma_I^k L_j} \phi(L_1, \dots, L_k) \Pi_I^k dL_j = (\Pi_I^k \Gamma_p(a_j) / \Gamma_p(a)) |Z|^{a - \frac{1}{2}(p+1)} |I-Z|^{b - \frac{1}{2}(p+1)}$$

where  $a = \Sigma_I^k a_j$ .

*Lemma 2.3*

If  $\phi$  is defined as in Lemma 2.2 then (de Waal (1970))

$$2.4 \quad \int_{0 < L_j < I, 0 < \Sigma_I^k L_j < I} \Pi_I^k |L_j|^{a_j - \frac{1}{2}(p+1)} |I - \Sigma_I^k L_j|^{b - \frac{1}{2}(p+1)} \\ \cdot C_\kappa(R \Sigma_I^k L_j) \Pi_I^k dL_j \\ = (\Gamma_p(b) \Pi_I^k \Gamma_p(a_j) / \Gamma_p(a+b)) / ((a)_\kappa / (a+b)_\kappa) C_\kappa(R)$$

for  $R$  a positive definite symmetric matrix.

*Lemma 2.4*

For  $\phi$  defined as before

$$2.5 \quad \int_{Z=\Sigma_I^k L_j} \phi(L_1, \dots, L_k) C_\kappa(R(I - \Sigma_I^k L_j)) \Pi_I^k dL_j \\ = (\Pi_I^k \Gamma_p(a_j) / \Gamma_p(a)) |Z|^{a - \frac{1}{2}(p+1)} |I-Z|^{b - \frac{1}{2}(p+1)} C_\kappa(R(I-Z))$$

where  $R > 0$ , symmetric, and  $\Sigma a_j = a$ .

*Proof*

Follows directly from Lemma 2.2.

*Lemma 2.5*

For  $\phi$  and  $R$  defined as before

$$2.6 \quad \int_{0 < L_j < I, 0 < \Sigma_I^k L_j < I} \Pi_I^k |L_j|^{a_j - \frac{1}{2}(p+1)} |I - \Sigma_I^k L_j|^{b - \frac{1}{2}(p+1)} C_\kappa(R(I - \Sigma_I^k L_j)) \Pi_I^k dL_j \\ = (\Pi_I^k \Gamma_p(a_j) / \Gamma_p(b) / \Gamma_p(a+b)) / ((b)_\kappa / (a+b)_\kappa) C_\kappa(R)$$

with  $a = \Sigma_I^k a_j$ .

*Proof*

The integral (2.6) is

$$\begin{aligned} & \int_{0 < Z < 1} \left( \int_{\Sigma L_j} Z \phi(L_1, \dots, L_k) \prod_1^k dL_j \right) C_K(R(1-Z)) dZ \\ &= \left( \prod_1^k \Gamma_p(a_j) / \Gamma_p(a) \right) \int_{0 < Z < 1} |Z|^{a-\frac{1}{2}(p+1)} |1-Z|^{b-\frac{1}{2}(p+1)} C_K(R(1-Z)) dZ \end{aligned}$$

Let  $W=1-Z$ ,  $J(Z \rightarrow W)=1$  and then using (2.2) yields the result.

*Lemma 2.6*

For  $R > 0$ , and symmetric

$$\begin{aligned} (2.7) \quad & \int_{V > 0} |V|^{a-\frac{1}{2}(p+1)} |1+V|^{-(a+b)} C_K(R(1+V)^{-1}) dV \\ &= (\Gamma_p(a) \Gamma_p(b) / \Gamma_p(a+b)) ((b)_K / (a+b)_K) C_K(R) \end{aligned}$$

*Proof*

$$\text{Let } Z = (1+V)^{-1} V, J(V \rightarrow Z) = |1-Z|^{-(p+1)}$$

and hence (2.7) is equivalent to

$$\int_{0 < Z < 1} |Z|^{a-\frac{1}{2}(p+1)} |1-Z|^{b-\frac{1}{2}(p+1)} C_K(R(1-Z)) dZ.$$

Using (2.2) yields the result.

*Lemma 2.7*

$$\text{If } \psi(V_1, \dots, V_k) = \prod_1^k |V_j|^{a_j-\frac{1}{2}(p+1)} |1+\sum_1^k V_j|^{-(a+b)}$$

then (Tan (1969) )

$$\begin{aligned} (2.8) \quad & \int_{\sum_1^k V_j = W} \psi(V_1, \dots, V_k) \prod_1^k dV_j \\ &= \left( \prod_1^k \Gamma_p(a_j) / \Gamma_p(a) \right) |W|^{a-\frac{1}{2}(p+1)} |1+W|^{-(a+b)}, \quad a = \sum_1^k a_j. \end{aligned}$$

*Lemma 2.8*

For  $\psi$  defined as above

$$\begin{aligned} (2.9) \quad & \int_{\sum_1^k V_j = W} \psi(V_1, \dots, V_k) C_K(R(1+\sum_1^k V_j)^{-1}) \prod_1^k dV_j \\ &= \left( \prod_1^k \Gamma_p(a_j) / \Gamma_p(a) \right) |W|^{a-\frac{1}{2}(p+1)} |1+W|^{-(a+b)} C_K(R(1+W)^{-1}) \end{aligned}$$

*Proof* Follows directly from Lemma 2.7.

*Lemma 2.9*

For  $\psi$  defined as above

$$\begin{aligned}
 (2.10) \quad \int \dots \int_{V_j > 0} \psi(V_1, \dots, V_k) C_k(R(I + \sum_I^k V_j)^{-I}) \Pi_I^k dV_j \\
 = (\Pi_I^k \Gamma_p(a_j) \Gamma_p(b) / \Gamma_p(a+b)) ((b)_k / (a+b)_k) C_k(R) \\
 \text{with } a = \sum_I^k a_j.
 \end{aligned}$$

*Proof*

The lemma can easily be established from either (2.9) or (2.6).

*Lemma 2.10*

If  $f(A)dA$  is the probability density of a symmetric matrix  $A$  ( $p \times p$ ) then the distribution of the diagonal matrix  $F$  of the characteristic roots of  $A$ ,  $A = HFH'$ , with  $H$  orthogonal is (James (1964))

$$2.11 \quad \int_{0(p)} f(HFH') (dH) (\Pi^{1/2} p^2 / \Gamma_p(1/2 p)) \alpha_p(F) df_1, \dots, df_p$$

$$F = \text{diag}(f_1, \dots, f_p) \text{ and } \alpha_p(F) = \sum_{i > j} (f_i - f_j)$$

*Lemma 2.11*

For symmetric  $S$  and  $T$  (James (1964))

$$\begin{aligned}
 2.12 \quad p^F q^F(a_1, \dots, a_p; b_1, \dots, b_q; S, T) \\
 = \int_{0(p)} p^F q^F(a_1, \dots, a_p; b_1, \dots, b_q; SHTH') (dH)
 \end{aligned}$$

and where the hypergeometric functions have been defined by James (1964).

*Lemma 2.12 (Khatri and Pillai (1968)).*

If  $A_k$  denotes the integral

$$2.13 \quad \int_D |Z|^{1/2(f-q-1)} \alpha_q(Z) C_k(Z) dz_1, \dots, dz_{q-1}$$

where

$$Z = \text{diag}(z_1, \dots, z_q), z_q = 1 - z_1 - \dots - z_{q-1}$$

and  $D$  is given by

$$2.14 \quad D: (0 < z_1 < z_2 < \dots < z_{q-2} < z_{q-1} < z_q = 1 - z_1 - z_2 - \dots - z_{q-1}),$$

then

$$2.15 \quad A_{\kappa} = ((\frac{1}{2}f)_{\kappa} \Gamma_q(\frac{1}{2}f) \Gamma_q(\frac{1}{2}q) C_{\kappa}(I_q) / \Pi^{\frac{1}{2}q^2} (\frac{1}{2}fq)_{\kappa} \Gamma(\frac{1}{2}fq)).$$

Lemma 2.13 (Troskie (1971a))

For symmetric matrix  $S$

$$2.16 \quad |I+S|^{-a} (a)_{\kappa} C_{\kappa} ((I+S)^{-1}) \\ = \sum_{m=0}^{\infty} \sum_{\eta} \sum_{\delta} ((-1)^m (a)_{\delta} g_{\kappa \eta}^{\delta} C_{\eta}(S) C_{\delta}(I) / C_{\eta}(1) m! )$$

where  $\delta = (\delta_1, \dots, \delta_q)$ ,  $\delta_1 \geq \delta_2 \geq \dots \geq \delta_q \geq 0$  and  $\sum \delta_i = m+k = d$ .

Lemma 2.14

For symmetric matrix  $S$  ( $p \times p$ )

$$2.17 \quad |I+S|^{-a} (a)_{\kappa} C_{\kappa}(S(I+S)^{-1}) \\ = k! \sum_{d=k}^{\infty} \sum_{\delta} (-1)^{k+d} (a)_{\delta} a_{\delta, \kappa} C_{\delta}(S) / d!$$

where  $a_{\delta, \kappa}$  is defined and tabulated by Constantine (1966) as

$$C_{\kappa}(I+A) / C_{\kappa}(I) = \sum_{n=0}^k \sum_{\eta} a_{\kappa, \eta} C_{\eta}(A) / C_{\eta}(I).$$

Proof

The lemma can be easily established by noting

$$2.18 \quad |I+S|^{-a} \sum_{k=0}^{\infty} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(S(I+S)^{-1}) C_{\kappa}(\theta) / k! C_{\delta}(I) \\ = \int_{0(p)} |I+H'(1-\theta)H|^{-a} dH$$

Theorem 2.1

If  $(L_1, \dots, L_k) \sim MD_I(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \Omega)$  then the density of

$$2.19 \quad L = \text{tr} \sum_{j=1}^k L_j$$

is given by

$$(2.20) \quad \Gamma_p(\frac{1}{2}m+n)/\Gamma_p(\frac{1}{2}n)\Gamma(\frac{1}{2}mp) \text{etr}(-\frac{1}{2}\Omega)$$

$$\begin{aligned} & \Sigma_k^\infty \Sigma_\kappa \Sigma_t^\infty = \Sigma_\tau \Sigma_j^k = \Sigma_\eta \Sigma_\delta ((-1)^j a_{\kappa, \eta} g_{\tau, \eta}^\delta \\ & \cdot (\frac{1}{2}(p+1-n))_\tau L^{\frac{1}{2}mp+d-1} (\frac{1}{2}m)_\delta C_\kappa(\frac{1}{2}\Omega) C_\delta(I) / (\frac{1}{2}mp)_d (\frac{1}{2}n)_\kappa C_\eta(I) k! t! \end{aligned}$$

for  $0 < L < 1$ , and where  $\delta = (\delta_1, \dots, \delta_p)$ ,  $\delta_1 \geq \delta_2 \geq \dots \geq \delta_p \geq 0$  and  $\Sigma \delta_i = d = t+j$ ;  $g_{\tau, \eta}^\delta$  is the coefficient of  $C_\delta(F)$  in the expansion  $C_\tau(F) C_\eta(F) = \Sigma_\delta g_{\tau, \eta}^\delta C_\delta(F)$  and  $a_{\kappa, \eta}$  is defined as above.

*Proof*

In the density of  $(L_1, \dots, L_k)$  given by (1.8) let  $Z = \Sigma_1^k L_j$ . Then using Lemma 2.4 we have the density of  $Z$  as

$$\begin{aligned} & (1/\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m)) |2\Sigma|^{-\frac{1}{2}(m+n)} \text{etr}(-\frac{1}{2}\Omega) |Z|^{\frac{1}{2}(m-p-1)} ||-Z|^{\frac{1}{2}(n-p-1)} \\ & \int_{T>0} \text{etr}(-\frac{1}{2}\Sigma^{-1}T) |T|^{\frac{1}{2}(m+n-p-1)} {}_0F_1(\frac{1}{2}n; \frac{1}{2}T^{\frac{1}{2}}\Omega \Sigma^{-1}T^{\frac{1}{2}}(I-Z))dT \end{aligned}$$

Let  $Z = HFH'$  with  $H$  orthogonal and where  $F = \text{diag}(f_1, \dots, f_p)$  ( $0 \leq f_1 \leq \dots \leq f_p \leq 1$ ) are the characteristic roots of  $Z$ . Then from Lemma 2.10 we get the density of  $F$  as

$$\begin{aligned} & (1/\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}m)) |2\Sigma|^{-\frac{1}{2}(m+n)} \text{etr}(-\frac{1}{2}\Omega) \int_{0(p)} |F|^{\frac{1}{2}(m-p-1)} \\ & ||-F|^{\frac{1}{2}(n-p-1)} \alpha_p(F) \int_{T>0} \text{etr}(-\frac{1}{2}\Sigma^{-1}T) |T|^{\frac{1}{2}(m+n-p-1)} \\ & {}_0F_1(\frac{1}{2}n; \frac{1}{2}T^{\frac{1}{2}}\Omega \Sigma^{-1}T^{\frac{1}{2}}H(1-F)H')dTdH. \end{aligned}$$

Integrating first with respect to  $H$  using Lemma 2.11 and then with respect to  $T$  using Lemma 2.1 yields the density of  $F$  as

$$\begin{aligned} (2.21) \quad & C(p, \frac{1}{2}m, \frac{1}{2}n) \text{etr}(-\frac{1}{2}\Omega) |F|^{\frac{1}{2}(m-p-1)} ||-F|^{\frac{1}{2}(n-p-1)} \alpha_p(F) \\ & \cdot {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega, 1-F), 0 \leq f_1 \leq \dots \leq f_p \leq 1 \end{aligned}$$

where

$$(2.22) \quad C(p, \frac{1}{2}m, \frac{1}{2}n) = \Pi^{\frac{1}{2}p^2} \Gamma_p(\frac{1}{2}(m+n))/\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}p).$$

Now using (see also Troskie and Money (1971))

$$(2.23) \quad |I-F|^{\frac{1}{2}(n-p-1)} C_K(I-F) = \sum_{t=0}^{\infty} \sum_{\tau} \sum_{j=0}^k \sum_{\eta} ((-1)^j \Sigma_{\delta} a_{K,\eta} b_{\tau,\eta}^{\delta} C_{\delta}(F) \cdot C_K(I)/C_{\eta}(I))$$

where

$$\Sigma_{\delta_j} = d = t+j, C_K(I-F) = \sum_{j=0}^k \sum_{\eta} (-1)^j C_{\eta}(F) C_K(I)/C_{\eta}(I)$$

and

$$C_{\tau}(F) C_{\eta}(F) = \Sigma_{\delta} g_{\tau,\eta}^{\delta} C_{\delta}(F),$$

in (2.21) and integrating  $(f_1, \dots, f_p)$  over the surface  $L = \text{tr } F = \sum_{i=1}^p f_i$ , using Lemma 2.12 yields the result.

### Corollary 2.1

If  $\Omega = 0$  then the central density of  $L \equiv \text{tr } \Sigma_1^k L_j$  is

$$(2.24) \quad \Gamma_p(\frac{1}{2}(m+n)) / \Gamma_p(\frac{1}{2}n) \Gamma(\frac{1}{2}mp) L^{\frac{1}{2}np-1} \sum_{t=0}^{\infty} \sum_{\tau} (\frac{1}{2}(p+1-n))_{\tau} L^t (\frac{1}{2}m)_{\tau} C_{\tau}(L) / ((\frac{1}{2}mp)_t t!) \quad \text{for } 0 < L < 1.$$

To derive the density of  $V = \text{tr } \Sigma_1^k L_j (I - \Sigma_1^k L_j)^{-1}$  note that if  $V_j$  is defined by (1.6) then  $V = \text{tr } \Sigma_1^k V_j$ .

### Theorem 2.2

If  $(L_1, \dots, L_k) \sim MD_p(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \Omega)$  then the density of

$$(2.25) \quad V = \text{tr } \Sigma_1^k L_j / (I - \Sigma_1^k L_j) = \text{tr } \Sigma_1^k V_j$$

where  $V_j$  is defined by (1.6) is given by

$$(2.26) \quad (\Gamma_p(\frac{1}{2}(m+n)) / \Gamma_p(\frac{1}{2}n) \Gamma(\frac{1}{2}mp)) \exp(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_{\kappa} \sum_{t=0}^{\infty} \sum_{\eta} \Sigma_{\delta} ((-1)^t (\frac{1}{2}(m+n))_{\delta} \cdot g_{\kappa,\eta}^{\delta} V^{\frac{1}{2}m(p+1)-1} C_K(\frac{1}{2}\Omega) C_{\delta}(I) / (\frac{1}{2}m)_{\eta} / (\frac{1}{2}mp)_t (\frac{1}{2}n)_{\kappa} C_K(I) k! t!), |V| < 1,$$

where  $m = \sum_{j=1}^k m_k$ ,  $\delta = (\delta_1, \dots, \delta_p)$ ;  $\delta_1 \geq \delta_2 \geq \dots \geq \delta_p \geq 0$ ,  $\Sigma \delta_i = t+k = d$ .

*Proof*

From (1.6) follows that  $(V_1, \dots, V_k) \sim MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n, \Omega)$  and since  $\text{tr } \Sigma_1^k V_j$

is invariant with respect to linear transformations we may assume  $\Sigma = I$  and hence the density of  $(V_1, \dots, V_k)$  is given by (1.10). In (1.10) let  $Z = \sum_1^k V_j$  and using Lemma 2.8 the density of  $Z$  is

$$(\Gamma_p(\frac{1}{2}m+n)/\Gamma_p(\frac{1}{2}m)\Gamma_p(\frac{1}{2}n))|Z|^{\frac{1}{2}(m-p-1)}|I+Z|^{-\frac{1}{2}(m+n)} \text{etr}(-\frac{1}{2}\Omega) \\ {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega(I+Z)^{-1}), Z > 0.$$

Using Lemma 2.10 and then Lemma 2.11 we have the density of the roots  $F = \text{diag}(f_1, \dots, f_p)$ ,  $(Z = HFH')$  of  $Z$  as

$$(2.27) \ C(p, \frac{1}{2}m, \frac{1}{2}n) \text{etr}(-\frac{1}{2}\Omega) |F|^{\frac{1}{2}(m-p-1)} |I+F|^{-\frac{1}{2}(m+n)} \alpha_p(F) \\ {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega(I+F)^{-1}), 0 < f_1 < \dots < f_p < 0.$$

Now using (2.16) in (2.27) and then integrating over the surface  $\Sigma f_1 = V$ , using Lemma 2.12, yields the result.

The density of  $V = \Sigma \text{tr } V_j$  has also been derived by De Waal (1968) using Laplace Transforms.

*Corollary 2.2*

*If  $\Omega = 0$  we have the central density of  $V$  as*

$$(2.28) \ (\Gamma_p(\frac{1}{2}(m+n))/\Gamma_p(\frac{1}{2}n)\Gamma(\frac{1}{2}mp))\Sigma_{t=0}^{\infty}\Sigma_{\eta}((-1)^t(\frac{1}{2}(m+n))_{\eta}(\frac{1}{2}m)_{\eta}V^{\frac{1}{2}mp+t-1} \\ \cdot C_{\eta}(I)/(\frac{1}{2}mp)_t t! ) \text{ for } |V| < 1.$$

*Theorem 2.3*

*If  $(L_1, \dots, L_k) \sim MD_1(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \Omega)$  then the density of*

$$(2.29) \ Q = \text{tr}(\Sigma_1^k L_j(I - \Sigma_1^k L_j)^{-1})^{-1} = \text{tr}(\Sigma_1^k V)^{-1}$$

*is*

$$(2.30) \ (\Gamma_p(\frac{1}{2}(m+n))/\Gamma_p(\frac{1}{2}m)\Gamma(\frac{1}{2}np) \text{etr}(-\frac{1}{2}\Omega) \Sigma_{k=0}^{\infty} \Sigma_{\kappa} \Sigma_{d=k}^{\infty} \Sigma_{\delta}((-1)^{k+d} C_{\kappa}(\frac{1}{2}\Omega) \\ \cdot Q^{\frac{1}{2}np+d-1}(\frac{1}{2}(m+n)) \delta^{a_{\delta, \kappa}}(\frac{1}{2}n) \delta C_{\delta}(I)/(\frac{1}{2}n)_{\kappa}(\frac{1}{2}np)_d C_{\kappa}(I)d! ), |Q| < 1, \text{ and}$$

$$\delta = (\delta_1, \dots, \delta_p), \delta_1 \geq \delta_2 \geq \dots \geq \delta_p \geq 0, \Sigma \delta_i = d.$$

*Proof*

From (1.6) follows that  $(V_1, \dots, V_k) \sim MD_2(\frac{1}{2}m_1, \dots, \frac{1}{2}m_k; \frac{1}{2}n; \Omega)$  and since  $Q$  is invariant with respect to linear transformations we may assume  $\Sigma = I$  and the density of  $(V_1, \dots, V_k)$  is again given by (1.10). Let  $Z = \Sigma_1^k V_j$  and let  $Y = Z^{-1}(J(Z)Y) = |Y|^{-(p+1)}$  then (using Lemma 2.8) the density of  $Y$  is given by

$$(2.31) \quad (\Gamma_p(\frac{1}{2}m+n)/\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n)) \text{etr}(-\frac{1}{2}\Omega) |Y|^{\frac{1}{2}(n-p-1)} |I+Y|^{-\frac{1}{2}(m+n)} \\ \cdot {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega Y (I+Y)^{-1}), Y > 0.$$

From Lemma 2.10 and Lemma 2.11 follows the density of the roots  $F = \text{diag}(f_1, \dots, f_p)$ ,  $Y = HFH'$ , of  $Y$  as

$$(2.32) \quad C(p, \frac{1}{2}m, \frac{1}{2}n) |F|^{\frac{1}{2}(n-p-1)} |I+F|^{-\frac{1}{2}(m+n)} \text{etr}(-\frac{1}{2}\Omega) \alpha_p(F) {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega; F(I+F)^{-1}), 0 < f_1 < \dots < f_p < \infty.$$

Using (2.17) in (2.32) and then integrating over the surface  $Q = \Sigma_{i=1}^p f_i$  using Lemma 2.12 yields the result.

To derive the distribution of  $|I - \Sigma_1^k L_j|$  we first consider its moments. Using Lemma 2.5 one can easily establish that the moments of  $|I - \Sigma_1^k L_j|$  are (see also De Waal (1968))

$$(2.33) \quad E(|I - \Sigma_1^k L_j|^h) = (\Gamma_p(\frac{1}{2}(m+n) - \Gamma_p(\frac{1}{2}n+h) / \Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}(m+n)+h) \\ \cdot \text{etr}(-\frac{1}{2}\Omega) {}_2F_2(\frac{1}{2}n+h, \frac{1}{2}(m+n); \frac{1}{2}n, \frac{1}{2}(m+n)+h; \frac{1}{2}\Omega), m = \Sigma_1^k m_j.$$

Notice that (2.33) is also the moments of  $I - L$  where  $L$  has a noncentral multivariate Beta distribution. Thus  $\Sigma_1^k L_j$  has a noncentral multivariate Beta distribution as was established in the proof of Theorem 2.1. Notice also that from (1.6) follows that  $|I - \Sigma_1^k L_j| = |I + \Sigma_1^k V_j|^{-1}$ . Hence (2.33) is also the moments of  $|I + \Sigma_1^k V_j|^{-1}$ .

The exact distribution of  $|I - \Sigma_1^k L_j|$  is complicated. Explicit expressions can only be given for the linear case, i.e. when  $\Omega$  is of rank one. The moments (2.33) then reduces to

$$(2.34) \quad E(|I - \Sigma_1^k L_j|^h) = (\Gamma_p(\frac{1}{2}(m+n) - \Gamma_p(\frac{1}{2}n+h) / \Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}(m+n)+h) \\ \cdot e^{-\frac{1}{2}\lambda^2} {}_2F_2(\frac{1}{2}n+h, \frac{1}{2}(m+n); \frac{1}{2}n, \frac{1}{2}(m+n)+h; \frac{1}{2}\lambda^2)$$

Using the well-known moments of Beta distribution it is easy to establish that (2.34) equals

$$(2.35) \quad E(|I - \sum_{j=1}^k L_j|)^h = E(\prod_{j=1}^p (1 - X_j)^h)$$

where  $X_j (j = 1, 2, \dots, p)$  has the density

$$(2.36) \quad (\Gamma(\frac{1}{2}(m+n+1-i))/\Gamma(\frac{1}{2}m)\Gamma(\frac{1}{2}(n+1-i)))x_i^{\frac{1}{2}m-1}(1-x_i)^{\frac{1}{2}(n+1-i)-1} \quad 0 < x_i < 1$$

and  $X_1$  has the density.

$$(2.37) \quad (\Gamma(\frac{1}{2}(m+n))/\Gamma(\frac{1}{2}m)\Gamma(\frac{1}{2}n))x_1^{\frac{1}{2}m-1}(1-x_1)^{\frac{1}{2}n-1}e^{-\frac{1}{2}\lambda^2} \\ \cdot {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\lambda^2(1-x_1)), 0 < x_1 < 1$$

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